

Cambridge IGCSE™

CHEMISTRY**0620/41**

Paper 4 Theory (Extended)

October/November 2025**MARK SCHEME**Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **13** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require <i>n</i> responses (e.g. State two reasons ...): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. • Incorrect responses should not be awarded credit but will still count towards <i>n</i>. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	information missing or insufficient for credit
	benefit of doubt given
	contradiction in response, mark not awarded
	incorrect point or mark not awarded
	key point attempted / working towards marking point / incomplete answer / response seen but not credited / blank page seen
	error carried forward applied
	incorrect or insufficient response, mark not awarded
	response has not answered question
	incorrect point or mark not awarded
	blank page seen

Annotation	Meaning
	correct point of mark awarded

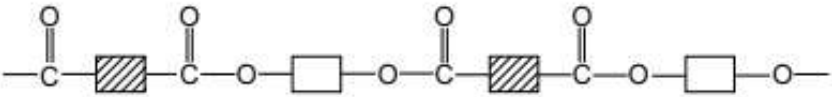
Question	Answer	Marks												
1(a)	<table border="1"> <tr> <td>particle</td><td>relative mass</td><td>relative charge</td></tr> <tr> <td>proton</td><td>1</td><td></td></tr> <tr> <td>neutron</td><td>1</td><td>0 / nil</td></tr> <tr> <td>electron</td><td></td><td>– 1</td></tr> </table> ✓(1) ✓(1)	particle	relative mass	relative charge	proton	1		neutron	1	0 / nil	electron		– 1	2
particle	relative mass	relative charge												
proton	1													
neutron	1	0 / nil												
electron		– 1												
1(b)(i)	M1 atoms of the same element that have the same number of protons M2 but different numbers of neutrons	2												
1(b)(ii)	they have the same number of electrons and therefore the same electronic configuration	1												
1(c)	M1 22 M2 16 and 18 M3 50 and 22 and to the left of the symbol M4 Ti M5 2+	5												
1(d)	nucleon number	1												
1(e)	M1 $\text{mol Ar} = 2.0 \div 40 = 0.05 \text{ (mol)}$ M2 number of atoms = $0.05 \times 6.02 \times 10^{23} = 3.01 \times 10^{22}$	2												

Question	Answer	Marks
2(a)	strontium / Sr	1
2(b)	M1 metallic M2 positive ions / cations M3 sea of electrons / mobile electrons / delocalised electrons M4 attraction between positive ions and electrons	4
2(c)(i)	M1 effervescence M2 solid dissolves M3 solution turns blue	3
2(c)(ii)	calcium hydroxide (1) hydrogen (1)	2
2(d)(i)	orange-red	1
2(d)(ii)	$2\text{Ca} + \text{O}_2 \rightarrow 2\text{CaO}$ M1 CaO as only product M2 correct equation	2
2(e)(i)	water(s) of crystallisation	1
2(e)(ii)	M1 $3.28 \div 164 = 0.02(00)$ M2 $1.44 \div 18 = 0.08(00)$ M3 $= \text{M2} \div \text{M1} = 4$	3

Question	Answer	Marks
3(a)	M1 2H ₂ O M2 state symbols (aq) and (l)	2
3(b)	neutralisation	1
3(c)	$0.0100 \times 1000 \div 40 = 0.25(0)$	1
3(d)	burette	1
3(e)	M1 mol H ₂ SO ₄ needed = $0.0100 \div 2 = 0.005(00)$ M2 vol of H ₂ SO ₄ needed = $0.005(00) \div 0.200 \times 1000 = 25 \text{ (cm}^3\text{)}$	2
3(f)	M1 yellow M2 orange	2
3(g)	repeat steps 1 and 3 without indicator	1
3(h)(i)	M1 (a solution containing the) maximum concentration of a solute dissolved in the solvent M2 at a specified temperature	2
3(h)(ii)	solubility decreases as temperature decreases	1
3(h)(iii)	none	1

Question	Answer	Marks
4(a)	M1 energy level of products below energy level of reactants M2 label of $\text{SO}_2(\text{g}) + 4\text{HF}(\text{g})$ on product line M3 upward arrow labelled E_a from energy level of reactants to the highest point of the curve M4 one downward arrow labelled ΔH AND energy change starting from energy level of reactants and finishing at energy level of products	4
4(b)	M1 $= (4 \times \text{S-F}) + (4 \times \text{O-H}) = (4 \times 330) + (4 \times 460) = 3160$ M2 bond energy of products $- 3160 = -54$ $= 3160 - (-54) = 3214$ M3 $(2 \times \text{S=O}) + (4 \times \text{H-F}) = 3214$ therefore $2 \times \text{S=O} = 3214 - (4 \times 570) = 934$ M4 $\text{S=O} = \text{M3} \div 2 = 934 \div 2 = 467$	4
4(c)	increased temperature: M1 (position of) equilibrium moves to left hand side M2 reaction is exothermic increased pressure: M3 (position of) equilibrium moves to left hand side M4 less (gaseous) moles on left hand side M5 no change / catalyst does not affect position of equilibrium	5
4(d)	M1 kinetic energy of particles decreases M2 frequency of collisions between particles decreases M3 lower percentage/proportion/fraction of collisions / particles have energy greater than / equal to activation energy	3

Question	Answer	Marks						
5(a)(i)	C _n H _{2n+1} OH	1						
5(a)(ii)	(same) functional group	1						
5(b)(i)	<table border="1"><tr><td>M1 25–35</td><td>M2 300</td></tr><tr><td>M3 yeast</td><td>M4 acid</td></tr><tr><td>M5 absence of oxygen</td><td>M6 60 atm</td></tr></table>	M1 25–35	M2 300	M3 yeast	M4 acid	M5 absence of oxygen	M6 60 atm	6
M1 25–35	M2 300							
M3 yeast	M4 acid							
M5 absence of oxygen	M6 60 atm							
5(b)(ii)	C ₆ H ₁₂ O ₆ → 2C ₂ H ₅ OH + 2CO ₂ M1 C ₆ H ₁₂ O ₆ as the only reactant on left M2 correct equation	2						
5(b)(iii)	C ₂ H ₄ + H ₂ O → C ₂ H ₅ OH M1 both C ₂ H ₄ + H ₂ O only on left M2 correct equation	2						
5(c)(i)	C ₄ H ₁₀ O ₂	1						
5(c)(ii)	2	1						
5(d)(i)	C ₂ H ₃ O ₂	1						
5(d)(ii)	hydrogen	1						

Question	Answer	Marks
5(e)(i)	 M1 any one correct fully displayed ester link between any two blocks M2 three fully displayed inter-block ester links with correct orientation M3 fully correct diagram showing two complete repeat units of polyester including continuation bonds	3
5(e)(ii)	condensation (polymerisation)	1